SIMATS
SAVEETHA INSTITUTE OF MEDICAL & TECHNICAL SCIENCES

CLOUD SECURITY FUNDAMENTALS



Protecting Data and Applications

OBJECTIVE Understand the models, shared responsibility, and controls that secure the cloud.

CONFIDENTIALITY

Only authorized users can access information.

INTEGRITY

Data remains accurate and unaltered.

AVAILABILITY

Systems and services remain accessible.

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Cloud Security Models

Choose the deployment model that matches risk, control, and business needs.

PUBLIC CLOUD

- Provider-operated infrastructure
- Scalable and cost-efficient
- Strong tenant isolation and configuration controls

PRIVATE CLOUD

- Dedicated to one organization
- Greater control and customization
- Organization manages more of the environment

HYBRID CLOUD

- Connects public and private systems
- Supports flexible workload placement
- Requires consistent identity and network policies

ZERO TRUST | Verify every access request • Enforce least privilege • Re-check continuously

Shared Responsibility Architecture

Security is shared—but the exact split depends on the cloud service model.

CLOUD PROVIDER

- Physical data centers and facilities
- Hardware, storage, and core networking
- Managed service platform components
- Availability and infrastructure maintenance

CUSTOMER / ORGANIZATION

- User identities, roles, and permissions
- Data classification, encryption choices, and backups
- Application code and secure configuration
- Compliance settings and incident response

IaaS • MORE CUSTOMER CONTROL

PaaS • SHARED PLATFORM

SaaS • PROVIDER-MANAGED APP

Key principle: the provider secures the cloud; the customer secures what they put in it and how they use it.

Real-World Security Practices

Layer controls to reduce risk, detect incidents early, and recover reliably.

01	**IDENTITY & ACCESS**	Use MFA, role-based access, and least privilege.
02	**DATA PROTECTION**	Encrypt data at rest and in transit; manage keys securely.
03	**NETWORK DEFENSE**	Use firewalls, private endpoints, segmentation, and secure rules.
04	**MONITOR & RESPOND**	Centralize logs, alert on unusual activity, and rehearse incident response.
05	**RESILIENCE**	Patch systems, test backups, and validate disaster recovery.

SECURE CLOUD = SHARED RESPONSIBILITY + DEFENSE IN DEPTH + CONTINUOUS IMPROVEMENT

Real-World Use Case

Illustrative scenario: a regional hospital moves patient records to a hybrid cloud.

01 THE CHALLENGE

- Patient records must stay private and available 24/7
- Regulations demand encryption and audit trails
- Legacy systems cannot scale for analytics

02 CONTROLS APPLIED

- MFA and role-based access for clinicians
- Encryption at rest and in transit; keys managed in-house
- Private endpoints and segmented networks
- Central logging with alerts on unusual access

03 THE OUTCOME

- Only authorized staff reach records (confidentiality)
- Data stays accurate and tamper-evident (integrity)
- Tested backups keep care running in an outage (availability)

HYBRID CLOUD + SHARED RESPONSIBILITY + LAYERED CONTROLS = PROTECTED PATIENT DATA

Conclusion & Key Takeaway

Cloud security works when responsibility is clear and protection is layered.

Security in the cloud is a shared job: know your side, layer your defenses, and keep verifying.

01 KNOW YOUR SIDE

The provider secures the cloud; you secure your identities, data, and configuration.

02 LAYER DEFENSES

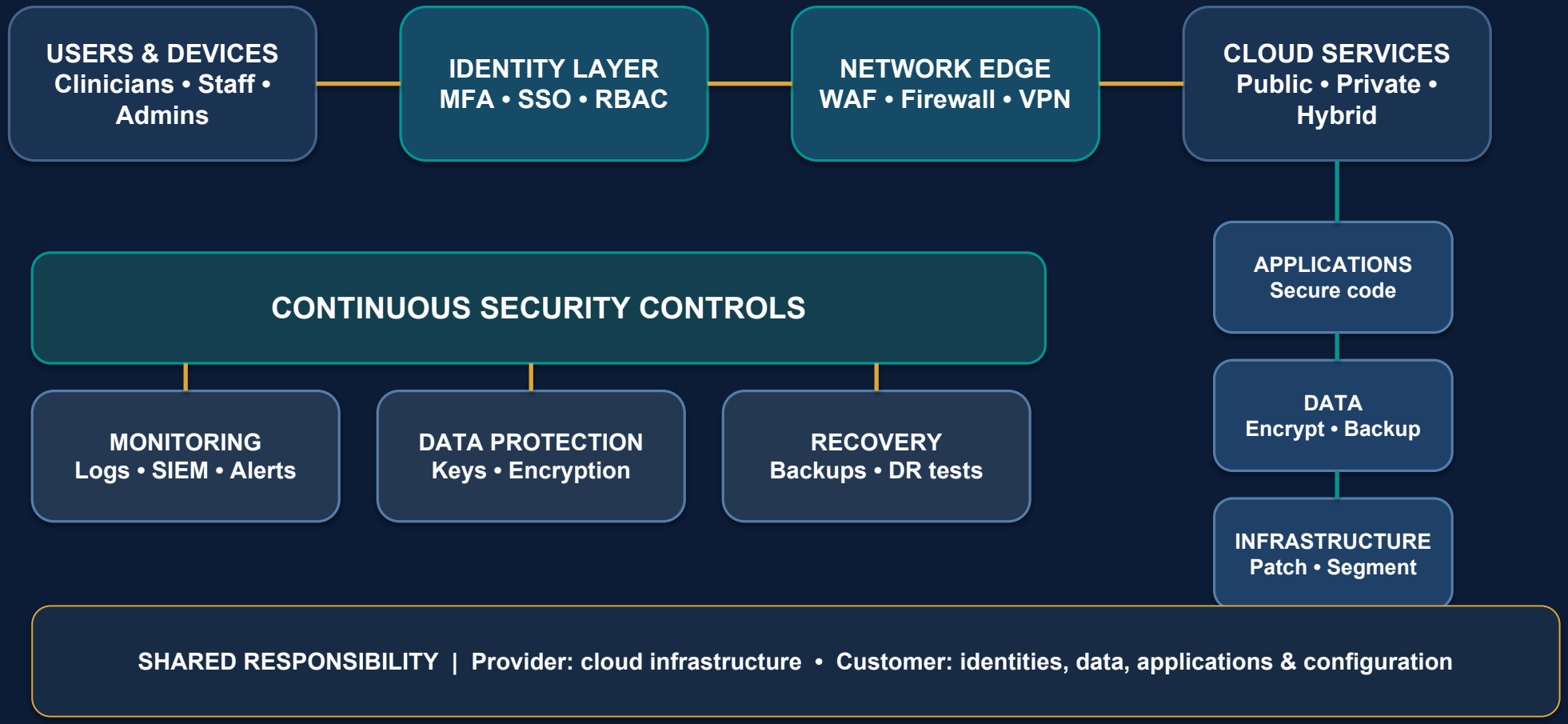
Combine identity, encryption, network controls, and monitoring so no single failure is fatal.

03 KEEP VERIFYING

Apply zero trust, test backups, and review access continuously.

CLOUD SECURITY ARCHITECTURE

Layered controls protect users, applications, data, and cloud infrastructure.



ZERO TRUST • LEAST PRIVILEGE • ENCRYPTION • MONITORING • RESILIENCE


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